BRIEF PRODUCT SPECIFICATION

nAD1001-18a

10-bit 1 MSPS Analog-to-Digital Converter IP

FEATURES

- TSMC CL018G 1.8 V Technology
- 6 to 12-bit Cyclic ADC
- 100 kSPS to 1 MSPS Conversion Rate
- 9 Muxed Single Ended Inputs
- Excellent Dynamic Performance
  58 dBFS SNR at Fin = 10 kHz
  65 dBc SFDR at Fin = 10 kHz
- Low Power Consumption
  2.9 mW at 1 MSPS
- Power Saving Idle Modes
- 0.21 mm^2 Core Area
- Internal Voltage Reference
- Single Ended Input Buffer

APPLICATIONS

- General Purpose Measurements
- Temperature Sensors
- Instrumentation
- Communication System Control Functions
  WLAN 802.11x / WiMAX 802.16x

GENERAL DESCRIPTION

The nAD1001-18a is a monolithic, low power, analog-to-digital converter silicon IP. It uses fully differential algorithmic architecture with digital error correction to provide 6 to 12 bit accuracy with 1 MSPS conversion speed. The core includes a sample-and-hold and an internal voltage reference that provides a nominal full-scale range of 1.2 V – it can be set between 0.8 V and 1.5 V using external references. It represents an ideal solution for general purpose measurements and temperature sensors.

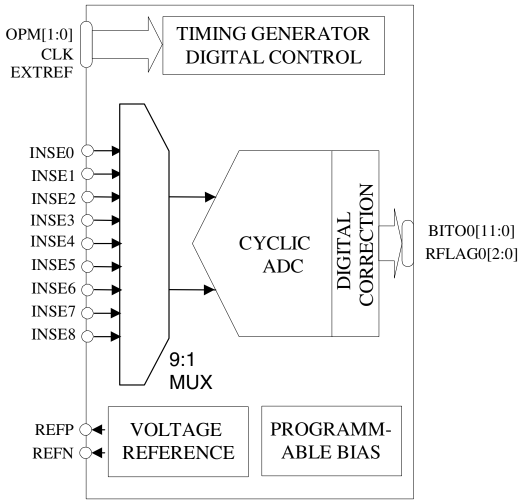

Figure 1. Functional block diagram

The ADC consumes only 2.9 mW at 1 MSPS operation. Combined with power saving idle modes the ADC is suitable for battery powered devices. Output data is available in a binary offset coded format. Three out-of-range indicator bits are also available for determining if the input signal is over-range, under-range or out-of-range.

Implemented in a generic 0.18 µm CMOS process, operating from a single 1.8 V supply and employing a differential architecture it represents an ideal ADC for highly integrated systems.

QUICK REFERENCE DATA

IP Type / Technology	Hard Macro / TSMC CL018G 1.8 V 1P5M			
IP Area / Dimensions	0.21 mm^2 / 0.514 × 0.4 mm			
Parameter	**Min.**	**Typ.**	**Max.**	**Unit**
Supply Voltage	1.6	1.8	2.0	V
Power Dissipation, 1 MSPS at F_{CLK} = 6 MHz		2.9		mW
Differential Non Linearity f_{IN}=0.9kHz			± 0.5	LSB
Integral Non Linearity f_{IN}=0.9kHz			± 0.75	LSB
Signal-to-Noise Ration, F_{IN} = 10 MHz		58		dB
Spurious-Free-Dynamic Range, F_{IN} = 10 MHz		65		dB

Table 1. nAD1001-18a quick reference data

ABOUT NORDIC SEMICONDUCTOR

Founded in 1983, Nordic Semiconductor ASA (OSE:NOD) is a Fabless Semiconductor specializing in short-range radio communication and high-speed data conversion. Nordic is head quartered in Trondheim, Norway,

MIXED SIGNAL IP OFFERINGS

Leading and emerging Fabless Semiconductor companies around the globe leverage Nordic's portfolio of off-the-shelf mixed signal intellectual property for the design of highly integrated circuits for broadband communication, digital imaging and video.

Our IP portfolio includes medium resolution (8 to 12-bit), high-speed (40 MSPS+) data converters (ADC and DAC) and low jitter clock generation (PLL) implemented in the latest deep submicron CMOS technologies. Featuring low power consumption, low-silicon area, and a minimum of required external components and technology options our products offers cost efficient integration of high-performance data converters in cost sensitive, high volume integrated circuit designs.

DELIVERABLES

Our mixed signal intellectual property is delivered as technology specific hard macros. Upon licensing we provide a complete design-kit and documentation that enables efficient, low risk, design-in and integration. The design kit for our off-the-shelf IPs includes:

- Full datasheet
- Evaluation board and samples
- Silicon validation report
- Integration Guidelines
- Physical design database (flat gds2)
- LVS Netlist (spice compatible)
- Footprint (.lef format)
- HDL model (verilog model)
- Timing model (.lib)
- Design-in and integration support

FURTHER INFORMATION

For further information about this product, such as full datasheet, silicon validation report, availability and licensing terms please contact us at: datacon@nordicsemi.no

DOCUMENT INFO	
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